nEEGlace

Assembly Guide for Preassembled PCBs | Step-by-Step Build Manual

4	22	10
PCBs to build	unique components	wiring steps

Boards in this kit:

BMS PCB	**Bela Cape PCB**	**Left MIC PCB**	**Right MIC PCB**
Battery Management	*Audio Processor*	*Left Microphone*	*Right Microphone*

!	**Note: Before you begin: This guide assumes basic soldering experience. Read every step fully before picking up your iron. A misoriented component can destroy the board.**

SECTION 1 — Tools and Safety

1.1 Tools Required

You will need all of the following before starting:

Soldering iron (temp-controlled, 320-350°C)	Fine-tip solder (0.5 mm or thinner, lead-free OK)
Flux pen (no-clean)	Solder wick/desoldering pump
Magnifying glass or USB microscope	Multimeter (voltage + continuity modes)
Isopropyl alcohol (IPA) + cotton swab for cleaning	ESD mat or anti-static wrist strap
Heat gun (200W)	Soldering iron paste
Tweezers	Soldering flux

TIP	TIP: Work on a bright, flat surface. A white sheet of paper under your board makes dropped SMD components much easier to find.

1.2 Safety Rules

!	**RULE 1: Always switch the BMS power switch to OFF before handling any wiring. Never connect the battery until Step 4.1.**

!	**RULE 2: SMD components are tiny and static-sensitive. Always wear an anti-static strap or regularly touch a grounded metal object.**

!	**RULE 3: Double-check component orientation BEFORE soldering. A backwards MOSFET or LDO can permanently damage the board.**

SECTION 2 — Component Identification Guide

Every component in this kit has an **ID number** (e.g. **[C1]**, **[R3]**). Learn to identify each one before you solder. This section is your visual dictionary.

2.1 Resistors

Resistors slow down or limit electrical current. Think of them as narrow pipes in a water system — they control how much electricity flows through.

ID	Part Name	What It Does	How to Identify It
R1	**10 kΩ Resistor x2 (MIC bias)**	Sets the bias voltage for the microphone circuit, ensures the mic gets the right operating conditions	*SMD, black rectangle ~1.6mm×0.8mm (0603). Marked '103' on top. Used on Bela Cape PCB.*
R2	**2.2 kΩ Resistor x2 (MIC bias)**	Limits current into the mic bias capacitors, prevents voltage spikes	*SMD, black rectangle ~1.6mm×0.8mm. Marked '222'. Used on Bela Cape PCB.*
R3	**1 MΩ Resistor (VD1 upper)**	The upper arm of the bio-signal voltage divider scales down the OpenBCI signal to safe levels	*SMD, marked '105'. Very high resistance keeps the circuit from loading the bio-signal. Bela Cape.*
R4	**10 kΩ Resistor (VD1 lower)**	Lower arm of VD1 divider, works with R3 to produce exactly 33mV output	*SMD, marked '103'. Goes next to R3 on the Bela Cape PCB.*
R5	**10 kΩ x2 (MIC gain)**	Sets the gain/amplification of the op-amp on each MIC PCB	*SMD, marked '103'. Two of these per MIC board. Part of the op-amp feedback loop.*
R6	**100 kΩ (MIC bias)**	Provides a high-impedance bias reference for the MEMS microphone	*SMD, marked '104'. One per MIC board. Slightly larger footprint than 10k.*
R7	**VD2 resistors**	Second voltage divider pair, conditions signal for another measurement channel	*Per schematic values, located near the VD2 label on the Bela Cape PCB.*
R8	**10 kΩ (BMS gate bias)**	Pulls the MOSFET gate, controls when charging protection activates	*SMD on BMS PCB. Must not be confused with 100kΩ, check the value before placing.*
R9	**100 kΩ (BMS gate bias)**	Second gate bias resistor, works with R8 to set MOSFET switching threshold	*SMD on BMS PCB. 10x higher value than R8, marking reads '104'.*

How to read SMD resistor codes:
SMD resistors use a 3-digit code: first two digits are the value, third is the multiplier (number of zeros to add).
103 = 10×10^3 = 10,000 Ω = 10 kΩ
104 = 10×10^4 = 100,000 Ω = 100 kΩ
105 = 10×10^5 = 1,000,000 Ω = 1 MΩ
222 = 22×10^2 = 2,200 Ω = 2.2 kΩ

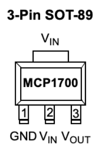
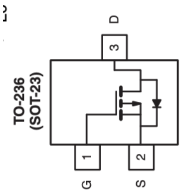
2.2 Capacitors

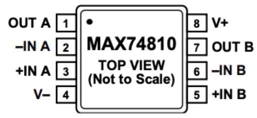
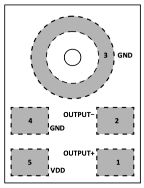
Capacitors store and release small bursts of electrical charge. They smooth out voltage wobbles, filter noise, and act as buffers between components.

ID	Part Name	What It Does	How to Identify It
C1	**10 µF Capacitor x2 (MIC bias caps)**	Smooths the MIC bias voltage; without these, the mic would hear its own power supply hum	*SMD, slightly taller than resistors. May be tan/beige colored. Bela Cape PCB positions C1, C2.*
C2	**1 µF Capacitor (BMS decoupling)**	Decouples the LDO output, filters fast voltage spikes that could destabilize the 3.3V rail	*SMD, smaller than 10µF caps. Marked '105' or just a dot. BMS PCB.*
C3	**10 µF Capacitor (BMS decoupling)**	Large decoupling cap absorbs slow power fluctuations from the battery	*SMD, same footprint as C1/C2 on Bela Cape. BMS PCB.*
C4	**10 µF x2 (MIC board decoupling)**	Power supply decoupling on each MIC PCB keeps the op-amp from oscillating	*Two per MIC board. Placed near op-amp power pins.*
C5	**10 µF x1 (MIC board bias)**	Additional decoupling for the 100kΩ mic bias resistor network	*One per MIC board. Silkscreen label confirms placement.*

Capacitor polarity warning:
Electrolytic capacitors (cylindrical, through-hole) have a + and − side. The LONGER leg is positive (+). The WHITE STRIPE on the body marks the negative (−) side.
The SMD ceramic capacitors in this kit (tan/beige blocks) are non-polar — you can solder them either way.

2.3 Active Components — The Smart Parts

ID	Part Name	What It Does	How to Identify It
U1	**LDO(MCP1700T)**	Low-Dropout Regulator. Converts the 5V from the HiLetgo module down to a clean, stable 3.3V for mics and OpenBCI. 'LDO' means it can regulate even when the input is only slightly above the output.	*SOT-89-3 package: flat tab on one side, 3 legs on the other. Legs = IN, GND, OUT.*  *Check the datasheet before soldering. On the BMS PCB.*
U2	**MOSFET (SSI2301CDS)**	The electronic switch disconnects the battery load when USB charging is connected. Protects the battery from being charged and discharged simultaneously.	*SOT-23-3P package: tiny, 3 legs. Legs = Gate (G), Source (S), Drain (D). ORIENTATION IS CRITICAL.*  *Dot or marking on package = pin 1 = Gate.*

U3	**Op-Amp (MIC PCB)(MAX74810)**	Operational Amplifier. Amplifies the tiny signal from the MEMS microphone to a usable audio level. Acts like a hearing aid for the circuit.	*SMD IC, 8-pin SOIC or TSSOP package. Pin 1 is marked with a dot or notch. Two on each MIC PCB.* 
MIC	**MEMS Mic(ICS-40638)**	Analog MEMS (Micro-Electro-Mechanical System) microphone. Converts sound waves into an electrical signal. Extremely tiny, handle with care.	*Small black square SMD component. Pin 1 is marked, alignment CRITICAL. One per MIC PCB.*  *This is the bottom view, since the terminals are placed on the bottom.*

!	**CRITICAL — Orientation matters on all ICs and MOSFETs. A single misaligned leg can destroy the component. Always verify pin 1 under magnification before applying solder.**

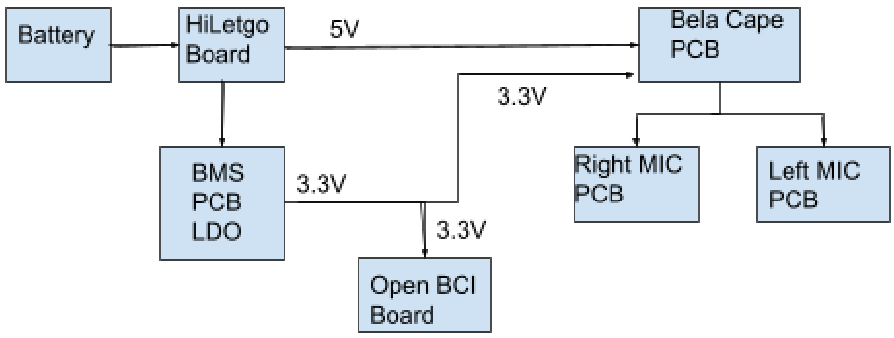
2.4 Connectors & Through-Hole Parts

ID	Part Name	What It Does	How to Identify It
J1	**2-pin Battery Connector**	Connects the LiPo battery to the BMS PCB. Red = positive (+), Black = negative (−). Never reverse this.	*JST PH 2.0mm or screw terminal (see your PCB). Located at 'Battery' label on BMS PCB.*
J2	**2-pin HiLetgo Connector**	Battery input to the HiLetgo charge+boost module.	*Matches J1 pitch. At 'HiLetgo' label on BMS PCB.*
J3	**2-pin HiLetgo 5V Out**	5V output from the HiLetgo module back to the BMS rail.	*At 'HiLetgo 5V' label on BMS PCB.*
J4	**2-pin Bela 5V**	Delivers 5V from BMS to the Bela Cape PCB.	*At 'Bela 5V' on BMS, connects to '5V Input' on Bela Cape.*
J5	**2-pin 3.3V OBCI**	Powers the OpenBCI board with 3.3V from the BMS LDO.	*At '3.3V OBCI' on BMS PCB.*
J6	**2-pin Mic 3.3V**	Supplies 3.3V mic bias rail from BMS to Bela Cape.	*At 'Mic 3.3V' on BMS, connects to 'MIC VD2' on Bela Cape.*
J7	**4-pin R-Mic Connector**	Carries 5V, 3.3V, GND, and audio signal to Right MIC PCB.	*At 'R-Mic' on Bela Cape. 4 pins, match all 4 carefully.*
J8	**4-pin L-Mic Connector**	Carries 5V, 3.3V, GND, and audio signal to Left MIC PCB.	*At 'L-Mic' on Bela Cape. 4 pins , mirror of J7.*
P1	**36-pin Header (BeagleBone P8)**	Connects Bela Cape to BeagleBone P8 expansion port , carries audio data and control signals.	*2×18 header, 2.54mm pitch. Long pins face down into BeagleBone.*
P2	**36-pin Header (BeagleBone P9)**	Connects Bela Cape to BeagleBone P9 expansion port , carries power and I2C signals.	*2×18 header, identical to P1. Pin 1 alignment critical.*

J9	**3.5mm Audio Jack**	Stereo TRS audio output , plug in headphones or recording interface here.	*Standard 3.5mm through-hole jack. Goes in the 'Audio Out' position on Bela Cape.*
SW1	**Power Switch**	Main system on/off switch. Always switch OFF before wiring. Always switch ON only after everything is connected.	*Through-hole slider or toggle switch. On BMS PCB.*
MOD 1	**HiLetgo Module**	Pre-built charge+boost module. Charges the LiPo battery via USB-C and boosts battery voltage to stable 5V. The heart of the power system.	*Small black PCB module. USB-C port visible. Mount onto J2 + J3 connectors.*

SECTION 3 : System Overview

SIGNAL FLOW DIAGRAM



BMS PCB	Battery Management System. Receives raw battery voltage, protects against overcharge via MOSFET [U2], boosts/regulates to 5V via HiLetgo module [MOD1], steps down to 3.3V via LDO [U1]. Has the power switch [SW1].
Bela Cape PCB	Audio processor cape for BeagleBone. Receives 5V + 3.3V, distributes mic power to both MIC PCBs, outputs stereo audio at 3.5mm jack [J9], sends bio-signals via VD1 divider to OpenBCI. Slots onto BeagleBone via headers [P1][P2].
MIC PCBs (×2)	Left and Right microphone boards. Each contains one MEMS mic [MIC], one op-amp [U3], and passive filter components. Powered by 5V + 3.3V from Bela Cape. Output goes back to Bela Cape as left/right stereo audio.

SECTION 4 — BMS PCB

This is Board **1 of 4**. It controls power for the entire system.

TIP	Look at Fig.1 (BMS PCB layout) and identify the labeled areas: Mic 3.3V, Bela 5V, Battery, HiLetgo, LDO, MOSFET, Switch, 3.3V OBCI, HiLetgo 5V.

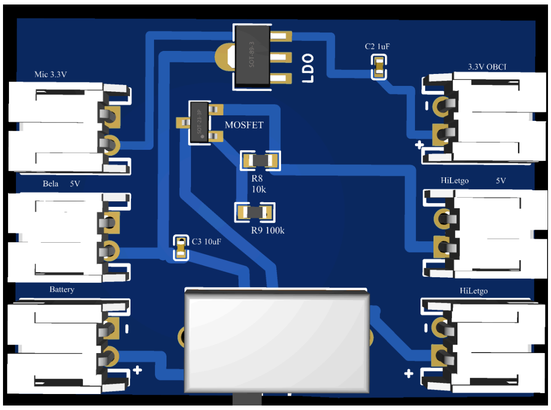


Fig.1: 3D view of BMS PCB

1. Verify that the preassembled BMS PCB looks exactly as in Fig.1.
2. Verify the values of gate-Bias Resistors [R8] and [R9] to be exactly 10kΩ and 100kΩ using a multimeter.
3. Verify the decoupling capacitors C2 and C3 using a multimeter. Set your digital multimeter to the Capacitance mode (indicated by the symbol(-||- or F) to read microfarads (µ F) or nanofarads (nF). If your multimeter lacks this setting, you can use the Resistance mode (Ω) as an alternative to test if the capacitor charges and discharges properly. If you are operating it in Resistance mode, if the measured value is more than 1MΩ, then the capacitor is working perfectly, and if it's less than 1MΩ or 0Ω, then the capacitor might be damaged.
4. Flip the board, check the switch pins from the back. Check that pins are fully through and contacts are not bridged. Confirm switch toggles between two positions with a satisfying click. Toggle to the OFF position before continuing.
5. Verify that all the connectors are soldered properly, flip the board, and check the connector pins from the back. Check that pins are fully through and contacts are not bridged.

4.1 Mount HiLetgo Module [MOD1]

Pre-built charge and boost module — no soldering required on the module itself

6. Solder red and black wires in the respective battery and 5V terminals labelled + and -.
7. Orient the HiLetgo module with USB-C port facing outward (away from board edge or toward accessible side per your enclosure plan).
8. Plug module's battery wires into [J2] — match polarity carefully (+/−).
9. Plug module's 5V output wires into [J3] — match polarity.
10. Confirm the module is secure. Do NOT power on yet.

BMS PCB — Assembly Checklist:

- R8 (10kΩ) and R9 (100kΩ) soldered and correct
- C2 (1µF) and C3 (10µF) soldered
- MOSFET [U2] orientation verified against schematic
- LDO [U1] orientation verified, tab soldered
- All 6 edge connectors J1–J6 soldered and labeled
- Switch [SW1] soldered and in the OFF position
- HiLetgo module [MOD1] plugged into J2 + J3

SECTION 5 — Bela Cape PCB Assembly

This is Board **2 of 4**. It is the audio heart of the system, running on a BeagleBone single-board computer. The PCB image shown in Fig.2 (3D view) is this board.

11. Verify that the preassembled Bela PCB looks exactly as in Fig.2.
12. Verify the values of MIC-Bias Resistors [R8] and [R9] to be exactly 2.2kΩ each using a multimeter.
13. Verify the MIC Bias capacitors C2 and C3 using a multimeter. Set your digital multimeter to the Capacitance mode (indicated by the symbol(-||- or F) to read microfarads (μ F) or nanofarads (nF). If your multimeter lacks this setting, you can use the Resistance mode (Ω) as an alternative to test if the capacitor charges and discharges properly. If you are operating it in Resistance mode, if the measured value is more than 1MΩ, then the capacitor is working perfectly, and if it's less than 1MΩ or 0Ω, then the capacitor might be damaged.
14. Verify the values of Voltage Divider Resistors [R3] and [R4] to be exactly 1MΩ and 10kΩ each using a multimeter.
15. Flip the board, check the Audio jack pins from the back. Check that pins are fully through and contacts are not bridged.
16. Verify that all the connectors are soldered properly, flip the board, and check the connector pins from the back. Check that pins are fully through and contacts are not bridged.

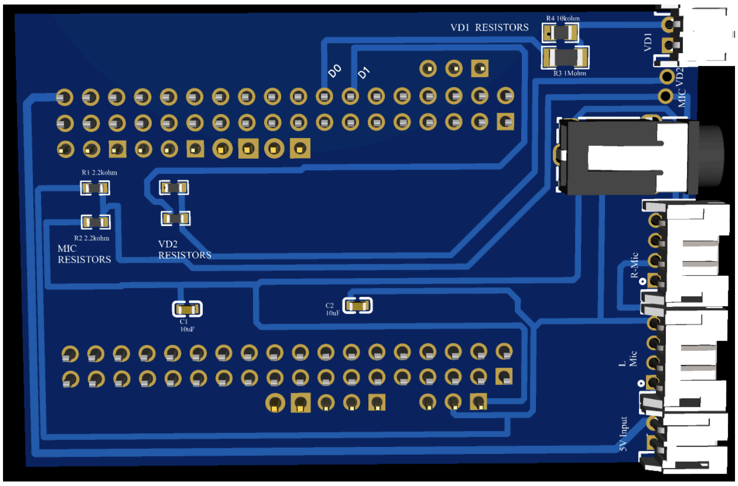


Fig.2: 3D view of Bela Cape

Bela Cape PCB — Assembly Checklist:

- R1, R2 (2.2kΩ each) soldered at MIC RESISTORS area
- C1, C2 (10μF each) soldered next to R1, R2
- R3 (1MΩ) upper VD1 pad, R4 (10kΩ) lower VD1 pad
- VD2 resistors per schematic
- All 6 edge connectors soldered and labeled correctly
- Audio jack [J9] seated flat and all pins soldered
- Headers P1 & P2 soldered with BeagleBone as alignment jig

SECTION 6 — Left MIC PCB & Right MIC PCB Assembly

These are Boards **3 and 4 of 4**. Both boards are **IDENTICAL**; Refer to Fig.3 for the 3D layout: you can see the Mic, Op-Amp, passives, and 4-pin output connector.

TIP	Fig.3 shows the 3D view of the MIC PCB. Locate: top-left = 10kΩ resistors, top-center = 100kΩ, left = 10µF caps, center = Op-Amp IC, bottom-left gray square = MEMS Microphone, right edge = 4-pin output connector.

17. Verify that the preassembled MIC PCBs look exactly as in Fig.3.
18. Verify the values of decoupling Resistors [R5a] and [R5b], [R6] to be exactly 10kΩ, 10kΩ, and 100kΩ, respectively, each using a multimeter.
19. Verify the decoupling capacitors C4a, C4b and C5 using a multimeter. Set your digital multimeter to the Capacitance mode (indicated by the symbol(-||- or F) to read microfarads (µ F) or nanofarads (nF). If your multimeter lacks this setting, you can use the Resistance mode (Ω) as an alternative to test if the capacitor charges and discharges properly. If you are operating it in Resistance mode, if the measured value is more than 1MΩ, then the capacitor is working perfectly, and if it's less than 1MΩ or 0Ω, then the capacitor might be damaged.
20. Flip the board, check the Audio jack pins from the back. Check that pins are fully through and contacts are not bridged.
21. Verify that all the connectors are soldered properly, flip the board, and check the connector pins from the back. Check that pins are fully through and contacts are not bridged.

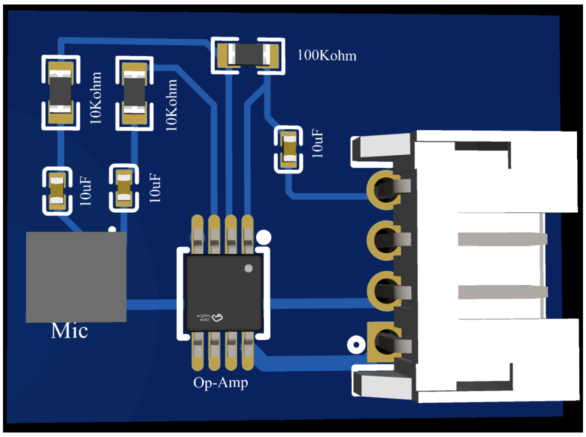


Fig.3: 3D view of the MEMS mic PCB

Both MIC PCBs — Checklist (do for EACH board):

- R5a and R5b (10kΩ each) soldered — values matched
- R6 (100kΩ) soldered at mic bias position
- C4a, C4b, C5 (10µF each) soldered
- Op-amp [U3] pin 1 aligned to PCB marker, all 8 legs soldered cleanly
- MEMS mic [MIC] pin 1 aligned, sound port facing outward
- 4-pin output connector soldered, pin order confirmed

SECTION 7 — System Wiring (Connecting All 5 Boards)

All four boards are now assembled. This section connects them together. **Work in the exact order shown. Do not connect the battery until Step 7.1 says to.**

!	**BEFORE STARTING: Confirm switch [SW1] on the BMS PCB is in the OFF position. Confirm battery is NOT connected to J1.**

7.1 Connect Battery to BMS [J1]

First wiring connection — still OFF

Connect the LiPo battery cable to the BMS connector [J1].

- **Red wire → (+) pin** of J1
- **Black wire → (−) pin** of J1
- Double-check polarity. Reversed battery connection = damaged BMS.
- Confirm switch [SW1] is OFF. Nothing should power on yet.

7.2 Connect BMS to Bela Cape: 5V Power [J4 → 5V Input]

5V + GND to Bela Cape

Run a 2-pin cable from BMS [J4] labeled 'Bela 5V' to Bela Cape connector labeled '5V Input'.

- Pin 1 (5V) connects to 5V
- Pin 2 (GND) connects to GND
- Match wire colors if using pre-made jumper — red=5V, black=GND

7.3 Connect BMS to Bela Cape: Mic Bias [J6 → MIC VD2]

3.3V mic power

Run a 2-pin cable from BMS [J6] labeled 'Mic 3.3V' to Bela Cape connector labeled 'MIC VD2'.

- This rail powers the microphone bias networks on both MIC PCBs

7.4 Connect BMS to OpenBCI [J5 → OpenBCI 3.3V In]

Powers the brain-signal board

Run a 2-pin cable from BMS [J5] labeled '3.3V OBCI' to the OpenBCI board's 3.3V power input terminal.

- Confirm OpenBCI's power input polarity matches — OpenBCI documentation specifies pin order

7.5 Connect Bela Cape VD1 → OpenBCI Signal Input

Bio-signal input to OpenBCI

The VD1 divider (R3 + R4) outputs a 33mV scaled bio-signal. Wire it to OpenBCI's analog signal input.

- VD1 (+) terminal → OpenBCI channel
- VD1 (−) terminal → OpenBCI Bias

7.6 Connect Bela Cape → Right MIC PCB [R-Mic → J7]

4-wire connection — power and audio

Wire #	Bela Cape R-Mic	Right MIC PCB	Signal
1	5V	5V	5V Power
2	3.3V	3.3V	Mic bias
3	GND	GND	Ground
4	R (Right audio)	Right MIC out	Audio signal

7.7 Connect Bela Cape → Left MIC PCB [L-Mic → J8]

4-wire connection — mirror of Step 7.7

Repeat Step 7.7 exactly, substituting 'L-Mic' connector on Bela Cape, and 'Left MIC PCB' as destination. Pin mapping is identical — just the audio channel is Left instead of Right.

7.8 Seat Bela Cape onto BeagleBone [P1 + P2]

Final board-to-board mating

22. Locate pin 1 marker on BeagleBone P8 header — usually a square pad or a dot on silkscreen.
23. Align Bela Cape P1 header pin 1 with BeagleBone P8 pin 1. They must match.
24. Press firmly and evenly straight down. Do NOT rock sideways — bent pins cannot be re-straightened easily.
25. Repeat for P2 onto BeagleBone P9.
26. Confirm both connectors are fully seated — no daylight visible between connector bodies.

7.9 Connect Audio Output [J9]

Plug in headphones, amp, or recording interface

Plug a 3.5mm TRS cable into the audio jack [J9] on the Bela Cape. This is your stereo audio output. No polarity to worry about — just push it in until it clicks.

SECTION 8 — Power-On & Verification

Do not skip this section. Test each stage before moving to the next. If something is wrong, catching it early saves hours of debugging.

8.1 Initial Power-On

Flip [SW1] to ON for the first time

27. Confirm all wiring from Section 7 is complete.
28. Confirm switch [SW1] is currently OFF.
29. Flip the switch to ON.
30. Immediately check: does anything smell like burning? See any smoke? If yes — SWITCH OFF IMMEDIATELY and look for solder bridges on the BMS.
31. If no smell/smoke: continue to voltage checks.

8.2 Voltage Verification Table

Use a multimeter in DC voltage mode. Black probe to any GND point. Measure at each connector:

Test Point	Expected Voltage	Tolerance	If Wrong...
BMS J4 'Bela 5V' connector	5.0 V	± 0.2V	Check HiLetgo module, check J3 wiring
BMS J5 '3.3V OBCI' connector	3.3 V	± 0.1V	Check LDO [U1] orientation, check C2/C3
BMS J6 'Mic 3.3V' connector	3.3 V	± 0.1V	Same LDO check as above
Bela Cape '5V Input' connector	5.0 V	± 0.2V	Check cable from J4 to 5V Input
Bela Cape 'MIC VD2' connector	3.3 V	± 0.1V	Check cable from J6 to MIC VD2
Right MIC PCB 5V pin	5.0 V	± 0.2V	Check R-Mic 4-pin cable
Right MIC PCB 3.3V pin	3.3 V	± 0.1V	Check R-Mic 4-pin cable pin 2
Left MIC PCB 5V pin	5.0 V	± 0.2V	Check L-Mic 4-pin cable

8.3 Audio Verification

32. Connect Bela to your computer via USB or network. Open the Bela web IDE at bela.local.
33. Confirm Bela boots successfully, the status LED on BeagleBone should blink, and bela. local should respond in the browser.
34. Load the 'passthrough' example from Bela IDE. This routes the mic input directly to the audio output.
35. Click Run. Speak near the Left MIC PCB, you should hear your voice in the headphones on the Left channel.
36. Speak near the Right MIC PCB; you should hear your voice on the Right channel.
37. If one channel is silent, see Troubleshooting Section 9.
38. Open the Bela 'scope' tool. Both left and right channels should show waveforms when you make noise.

8.4 Charging Test

39. Connect a USB-C cable to the HiLetgo module [MOD1].
40. The system should power DOWN automatically. This is correct behavior, designed to prevent simultaneous charge+discharge.
41. Disconnect USB-C. The system should power back ON automatically within a few seconds.

TIP	If the system does NOT power off when the USB is connected, the MOSFET [U2] is likely backwards. Desolder and re-orient, checking against the schematic.

SECTION 9 — Troubleshooting Guide

Something not working? This table covers the most common problems and their fixes.

Symptom	Likely Cause	Fix / Check
No power anywhere, nothing lights up	Switch [SW1] off, dead battery, or battery connector [J1] reversed	Flip the switch ON. Check battery voltage (≥3.7V). Verify J1 polarity.
The system won't power on even with a fresh battery	MOSFET [U2] installed backwards	Desolder U2 and re-orient. Check gate (G), drain (D), and source (S) against the schematic.
The system never powers off when a USB-C is plugged in	MOSFET [U2] reversed, or R8 (10kΩ) and R9 (100kΩ) swapped	Desolder U2 and check orientation. Confirm R8='103' and R9='104'.
No 3.3V output at J5 or J6	LDO [U1] orientation wrong, or C2/C3 missing/bridged	Verify LDO IN-GND-OUT pin order against the MCP1700T datasheet. Check capacitors for solder bridges.
One MIC channel completely silent	Broken 4-pin cable, wrong pin order, or bad op-amp solder joint	Check cable continuity with a multimeter. Inspect op-amp [U3] under magnifier for lifted pins or bridges.
Both MIC channels silent	Missing 3.3V on MIC VD2, or C1/C2 missing on Bela Cape	Measure voltage at the MIC VD2 connector. Check the J6 cable. Inspect C1 and C2 on Bela Cape.
Audio sounds distorted or buzzy	R1/R2 (2.2kΩ) values wrong, or C1/C2 poor solder joints	Measure R1/R2 with a multimeter (remove power first). Re-flow C1/C2 solder joints.
OpenBCI receives no signal from VD1	R3/R4 swapped, or VD1 connector wired backwards	Verify R3=1MΩ (upper pad) and R4=10kΩ (lower pad). Check VD1 connector +/− polarity.
BeagleBone won't boot / no bela.local	Headers P1/P2 misaligned or not fully seated	Press the Bela Cape firmly down onto the BeagleBone. Look for bent pins on P8/P9.
MEMS mic makes crackle/static noise	Sound port blocked or C4/C5 missing	Check mic port for flux residue — clean gently with IPA and thin brush. Verify C4a, C4b, C5 present.

SECTION 10 — Quick Reference Card

Cut out or bookmark this page. It summarizes every component ID, its location, and critical orientation notes.

All Component IDs at a Glance

ID	Part	Value	Location	Critical Note
R1	Resistor	2.2 kΩ	Bela Cape, MIC RESISTORS	Non-polar, SMD
R2	Resistor	2.2 kΩ	Bela Cape, MIC RESISTORS	Non-polar, SMD
R3	Resistor	1 MΩ	Bela Cape, VD1 upper	Marked 105 — upper pad
R4	Resistor	10 kΩ	Bela Cape, VD1 lower	Marked 103 — lower pad
R5a/b	Resistor	10 kΩ ×2	MIC PCB	Op-amp gain — matched pair
R6	Resistor	100 kΩ	MIC PCB	Mic bias, marked 104
R7	Resistor	per schematic	Bela Cape	Check the schematic for values
R8	Resistor	10 kΩ	BMS PCB	Gate bias, marked 103
R9	Resistor	100 kΩ	BMS PCB	Gate bias, marked 104 — do not swap with R8
C1	Capacitor	10 µF	Bela Cape	Non-polar SMD ceramic
C2	Capacitor	1 µF	BMS PCB	Non-polar SMD ceramic
C3	Capacitor	10 µF	BMS PCB	Non-polar SMD ceramic
C4a/b	Capacitor	10 µF ×2	MIC PCB	Op-amp decoupling
C5	Capacitor	10 µF	MIC PCB	Mic bias decoupling
U1	LDO	MCP1700T-3302	BMS PCB	SOT-89-3: verify IN/GND/OUT
U2	MOSFET	SSI2301CDS-T1-E3	BMS PCB	SOT-23-3P: G/S/D — CRITICAL orientation
U3	Op-Amp	per BOM	MIC PCBs	8-pin SMD: pin 1 = dot/notch
MIC	MEMS Mic	MMICT390200012	MIC PCBs	Sound port faces outward — critical
SW1	Switch	Power	BMS PCB	OFF before any wiring
MOD1	HiLetgo	5V boost + charger	BMS PCB	USB-C for charging
P1	Header	2×18 / 36-pin	Bela Cape	BeagleBone P8 — use BB as jig
P2	Header	2×18 / 36-pin	Bela Cape	BeagleBone P9 — use BB as jig

Wiring Connections Summary

Step	From	To	Signal
7.1	Battery	BMS J1 [Battery]	Battery power
7.3	BMS J4 [Bela 5V]	Bela Cape [5V Input]	5V + GND
7.4	BMS J6 [Mic 3.3V]	Bela Cape [MIC VD2]	3.3V mic bias
7.5	BMS J5 [3.3V OBCI]	OpenBCI power input	3.3V + GND
7.6	Bela Cape [VD1]	OpenBCI signal input	33mV bio-signal
7.7	Bela Cape [R-Mic]	Right MIC PCB	5V, 3.3V, GND, audio
7.8	Bela Cape [L-Mic]	Left MIC PCB	5V, 3.3V, GND, audio
7.9	Bela Cape P1	BeagleBone P8	36-pin expansion
7.9	Bela Cape P2	BeagleBone P9	36-pin expansion

Assembly Complete!

You have built the nEEGlace.
For firmware, Bela programming, and signal processing — see your project documentation.